CPE 381 Classwork 10: Fourier Series

Student Name (as in Canvas): _____

A Number: _____

Points: 20 Duration: 15 minutes Due: September 30, 2026 (In-Class)

1. Find the Fourier coefficients for $x(t) = B + A\sin(\Omega_0 t + \theta)$ and hence write the fourier representation in terms of complex exponentials. **(10 Points)**

Answer

2. $x(t) = \cos(2t + \frac{\pi}{4})$. Find the Fourier series representation by finding the Fourier series coefficients. In order to do that, first, write down the value of fundamental frequency Ω_0. **(8 + 2 Points)**

Answer